



# **Stocker Online Requirement Document**

Version. 2.31

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# 1 Introduction

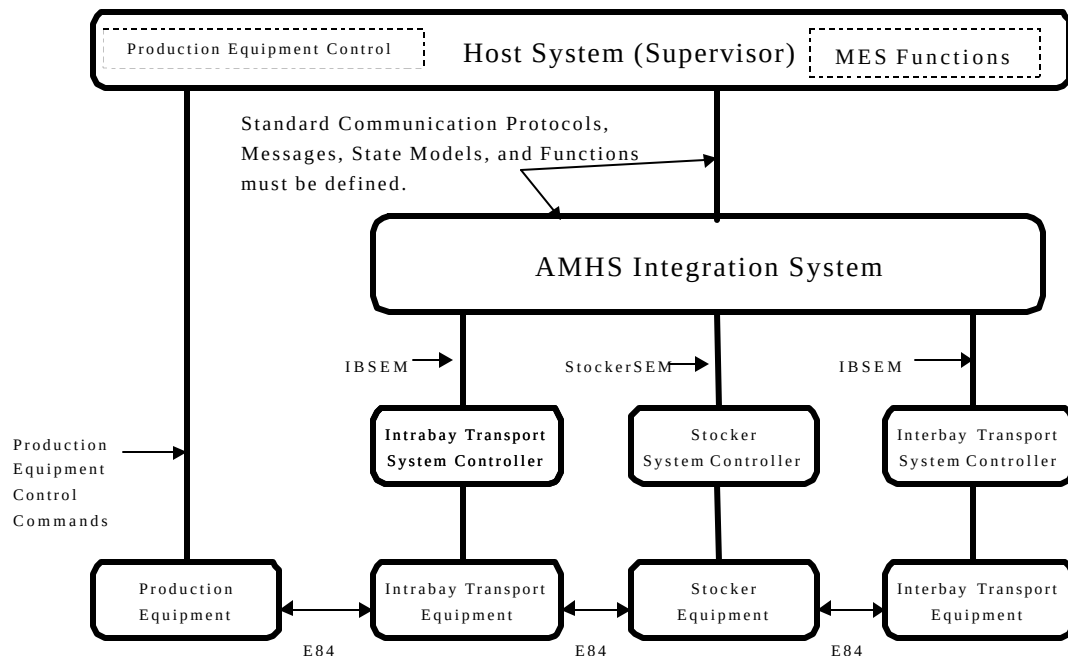
## 1.1 Objectives

This document specifies the Selete requirement for stocker online.

## 1.2 Back Ground

With 300mm semiconductor Fab line of the future, the fully automatic transport of the carrier in a line is demanded as a result of the request about the human engineering and the safety accompanied by the expansion of a wafer size, and the increase of the weight of 300mm wafer carrier. For this reason, AMHS equipment is required to be integrated organically and cost-effectively with a host system.

In order to attain this purpose, each AMHS equipment must use the common physical and logical interfaces so that AMHS equipment can be interoperable. In order to realize the logical interoperability, I300I and J300 proposed AMHS system configuration and AMHS integration system as shown in Figure 1 below at “6.2 AMHS framework” on CIM Global Joint Guidance. They also specified user requirement of AMHS integration system on “AMHS Framework User System requirement Document”.



**Figure 1 AMHS Framework**

The above-mentioned AMHS framework specifies that the stocker should use Stocker SEM (SEMI E88 Specification for AMHS Storage SEM) as a communication interface with a host system (AMHS integration system).

In order to realize the interoperable stocker, Selete has analyzed the above-mentioned "AMHS Framework User System Requirement Document" and "SEMI E88 Specification for AMHS Storage SEM (Stocker SEM)." The result of analysis reveals the following.

- 1) In order to realize interoperability of a communication interface, additional specification is needed for Stocker SEM
- 2) Selete's interpretations must be fixed about the ambiguous points of the description of Stocker SEM itself.

This specification is intended to realize the online specification of interoperable stocker and to specify the additional Selete specification to Stocker SEM and the Selete interpretation.

### 1.3 Reference Documents

*Selete AMHS Communications Scenario Document*

*Selete Intrabay/Interbay Transport System Online Requirement Document*

*Selete AMHS Communication Test Specification*

*Selete Integrator Requirement Document*

*I300I / J300 AMHS Framework User System Requirement Document*

*SEMI Doc E5 – SEMI Equipment Communications Standard2 Message Content (SECSII)*

*SEMI Doc E30 – Generic Model for Communications and Control of Manufacturing-Equipment(GEM)*

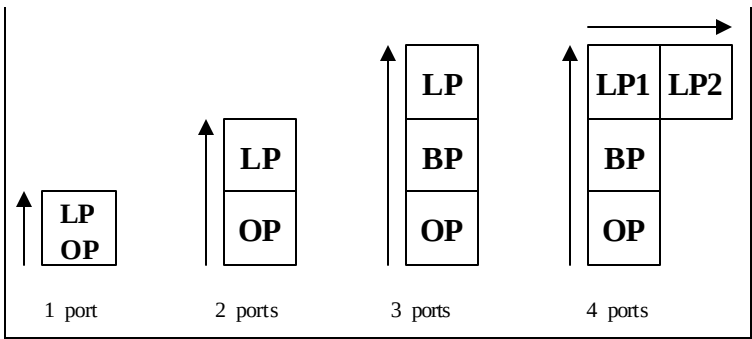
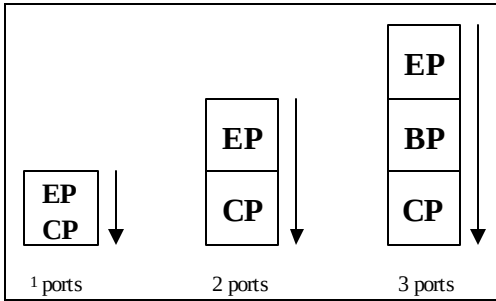
*SEMI Doc E37 – High-Speed SECS Message Services (HSMS) Generic Services*

*SEMI Doc E37.1 – HSMS Single Session Mode(HSMS-SS)*

*SEMI Doc E82 – Specification for Intrabay/Interbay AMHS SEM(IBSEM)*

*SEMI Doc E88 – Specification for AMHS Storage SEM (Stocker SEM)*

## 2 Terminology

|                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output Shuttle                       | <p>It consist of OP (Output shuttle), BP (Buffer Port), and LP (Loading Port). According to the placement of port, it becomes as it is shown in the following figure.</p>  <p>The diagram illustrates four configurations of an Output Shuttle based on the number of ports. Each configuration is shown within a rectangular frame with an upward-pointing arrow on the left side. <br/> 1 port: A single box containing 'LP' and 'OP'. <br/> 2 ports: Two stacked boxes, 'LP' on top and 'OP' on the bottom. <br/> 3 ports: Three stacked boxes, 'LP' on top, 'BP' in the middle, and 'OP' on the bottom. <br/> 4 ports: Four stacked boxes, 'LP1' and 'LP2' side-by-side at the top, 'BP' in the middle, and 'OP' at the bottom. A horizontal arrow points from 'LP1' to 'LP2'.</p> |
| Input shuttle                        | <p>It consists of EP (Entering Port), BP (Buffer Port), and CP (Crane accessible Port). According to the number of ports, it becomes as it is shown in the following figure..</p>  <p>The diagram illustrates three configurations of an Input Shuttle based on the number of ports. Each configuration is shown within a rectangular frame with a downward-pointing arrow on the left side. <br/> 1 ports: A single box containing 'EP' and 'CP'. <br/> 2 ports: Two stacked boxes, 'EP' on top and 'CP' on the bottom. <br/> 3 ports: Three stacked boxes, 'EP' on top, 'BP' in the middle, and 'CP' on the bottom.</p>                                                                                                                                                             |
| Remote Command                       | It refers to "remote command" that is specified by Stocker SEM. TRANSFER command is one of the remote commands.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Buffer Port<br>Crane Accessible Port | The location for the buffering at input and output of stocker.<br>The port location that can be accessed by the stocker crane at the input shuttle of stocker.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Entering Port                        | The port location that can be accessed by the operator or the vehicle at the input shuttle of stocker.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Loading Port                         | The port location that can be accessed by the operator or the vehicle at the output shuttle of stocker.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Output Port                          | The port location that can be accessed by the stocker crane at the output shuttle of stocker.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Transfer Port                        | The location on the transport system at which a change of equipment ownership of the carrier occurs. Transfer Posts in a stocker are Entering Port. (EP) and Loading Port (LP).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

### 3 Compliance to SEMI standards

#### 3.1 SECS communication

- SECS communication between a stocker controller (SC) and a host system must be based on the following SEMI standards.

| SEMI Standards |                                                                  |
|----------------|------------------------------------------------------------------|
| SEMI Doc E5    | SEMI Equipment Communications Standard2 Message Content (SECSII) |
| SEMI Doc E37   | High-Speed SECS Message Services(HSMS) Generic Services          |
| SEMI Doc E37.1 | HSMS Single Session Mode(HSMS-SS)                                |

**Table 1 SEMI standards related to SECS communication**

##### 3.1.1 HSMS CONNECT MODE

- HSMS Connect Mode is required to perform in both active and passive mode.

#### 3.2 Equipment Model

- A stocker controller (SC) must implement the equipment model based on the following SEMI standards.

| SEMI Standards |                                                                                                                                                                                                                                                              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEMI Doc E30   | The Standard of SEMI is specified in the following sections listed in the standard of Generic Equipment Model(GEM)<br>Chapter 1,2,3: All sections<br>Chapter 4: Section 4.1,4.2.1,4.2.2,4.2.5,4.2.6,4.3,4.5,4.7,4.8,4.9,4.10,4.12<br>Chapter 8: All sections |
| SEMI Doc E88   | Specification for AMHS Storage SEM (Stocker SEM)                                                                                                                                                                                                             |

**Table 2 SEMI standards related to Equipment Models**

[Note 1]

- The "14 GEM Capabilities" of Stocker SEM specifies GEM function specified by Stocker SEM. Refer to the table on the following page. When we show the description of performance that is marked by X, among the fundamental GEM requirements and the additional capabilities in the table on the following table, then the result will be as, shown in the above-mentioned table.

| GEM fundamental requirements      | Additional capabilities            |   |
|-----------------------------------|------------------------------------|---|
| State Model                       | Establish Communications           | X |
| Equipment Processing State        | Dynamic Event Report Configuration | X |
| Host-initiated S1F13/F14 Scenario | Variable Data Collection           |   |
| Event report                      | Trace Data Collection              |   |
| Online Confirmation               | Status Data Collection             | X |
| Error Message                     | Alarm Management                   | X |
| Control(operator-initiated)       | Remote Control                     |   |
| Documentation                     | Equipment Constants                | X |
|                                   | Process Program Management         |   |
|                                   | Material Movement                  | X |
|                                   | Equipment Terminal Service         | X |
|                                   | Clock                              | X |
|                                   | Limits Monitoring                  |   |
|                                   | Spooling                           |   |
|                                   | Control(host-initiated)            | X |

**Figure 2 GEM Capabilities Required by Stocker SEM**

### 3.2.1 State Model List

- The state models that SC should implement are described in both SEM and GEM, and since they are incomprehensible, they are listed here.
  - 1) Communication state model (refer to GEM 3.2 Communication State model)
  - 2) Control state model (refer to GEM 3.3 Control State model)
  - 3) SC state model (refer to SEM 8.2 SC State Model)
  - 4) Transfer command state model (refer to SEM 8.3 TRANSFER Command State Model)
  - 5) Carrier state model (refer to SEM 8.4 Stocker Carrier State Model)
  - 6) Crane state model (refer to SEM 8.5 Stocker Crane State Model)
- In addition to these, SC has the state model of HSMSSS  
(Refer to 5. "Summary and State Machine of HSMSSS" in "SEMI DOC.E37.1 HSMS-SS")

### 3.2.2 Entities that have State Models

- The entity that owns the above-mentioned state model is shown in the following figures. For example, the above-mentioned 3) SC state model has one model for one SC. The 5) Carrier state model has one model for each carrier within a stocker. In that sense, the entity that owns SC state model is SC, while the entity that owns a carrier state model is a carrier.

This shows how it should be seen from a host, and it is outside this specification whether or not it is implemented actually in this way.

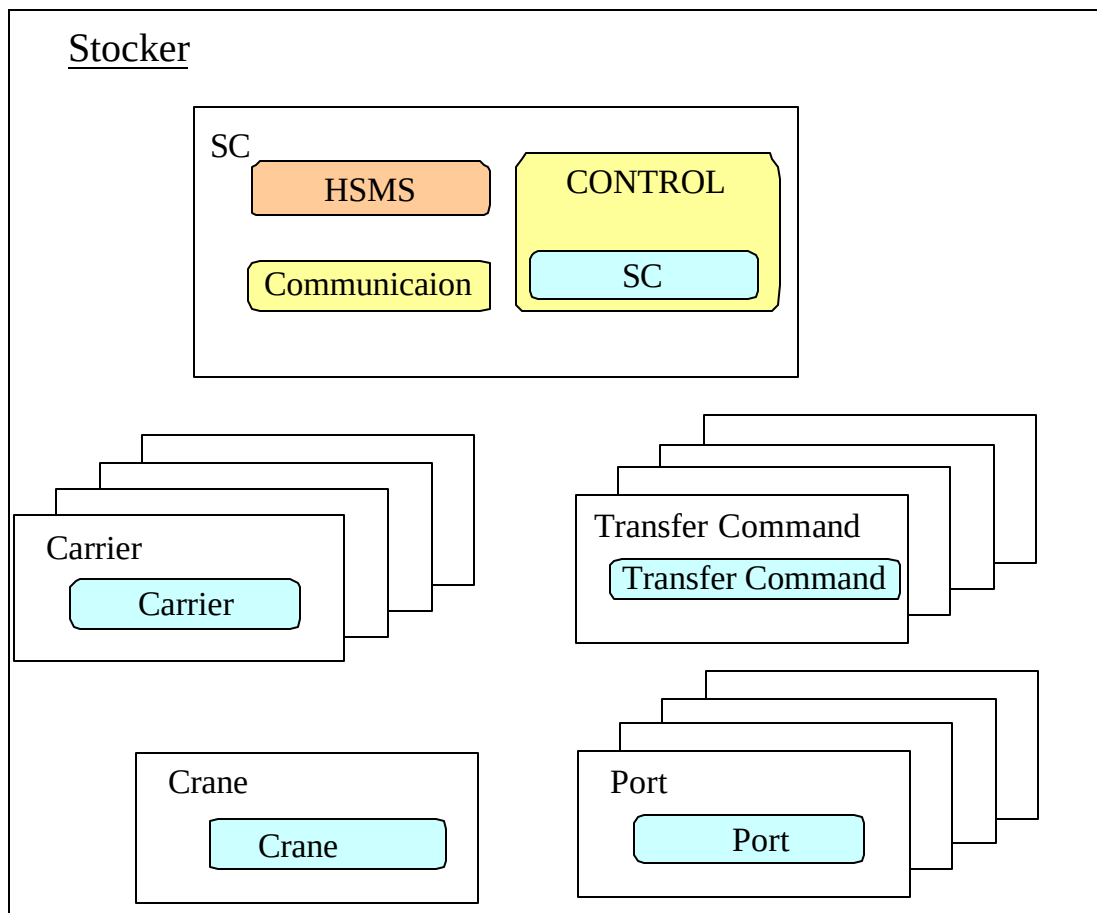


Figure 3 Entities that have State Model

## 4 Data Items

This chapter explains SECS message data item that is used by the stocker.

### 4.1 SECS Data Items

The data items of SECS message that are used by the stocker is shown in next table.

Item description of a table

Name : The name of a data item

Format : The format of the value of a data item

Size : The size of the value of a data item

Description : Description of a data item, or the range of data value

**Table 3 SECS Data Items used by Stocker**

| Name    | Format                       | Size  | Description                                                                                                                                                                                |
|---------|------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACKC5   | B                            | 1     | 0=Accepted >0=Error; Not accepted 1-63 Reserved                                                                                                                                            |
| ACKC6   | B                            | 1     | 0=Accepted >0= Error; Not accepted 1-63 Reserved                                                                                                                                           |
| ACKC10  | B                            | 1     | 0=Accepted 1=Messages will not be displayed<br>2=Terminal not available 3-63 Reserved                                                                                                      |
| ALCD    | B                            | 1     | bit8=1 Alarm Set<br>bit8=0 Alarm Cleared<br>bit1-7=0 Alarm not in use                                                                                                                      |
| ALED    | B                            | 1     | bit8=1 Enable Alarm bit8=0 Disable Alarm                                                                                                                                                   |
| ALID    | U4                           |       | Alarm identification                                                                                                                                                                       |
| ALTX    | A                            | Max40 | Alarm text limited to 40 characters                                                                                                                                                        |
| CEED    | BOOLEAN                      | 1     | False = Disable True = Enable                                                                                                                                                              |
| CEID    | U4                           |       | Collected Event ID                                                                                                                                                                         |
| CEPACK  | L,U1                         |       | 0=No Error<br>1=Parameter name (CPNAME) does not exist<br>2= Illegal value specified for CEPVAL<br>3=Illegal value specified for CEPVAL<br>4=Parameter name (CPNAME) not valid as used     |
| CEPVAL  | A,B,BOOLEAN,<br>Fx,Ix,J,L,Ux |       | Command Enhanced Parameter Value                                                                                                                                                           |
| COMMACK | B                            | 1     | 0=Accepted 1=Denied,Try Again 2-63 Reserved                                                                                                                                                |
| CPACK   | B                            | 1     | 0=Accepted<br>1= Parameter name (CPNAME) does not exist<br>2=Illegal value specified for CPVAL<br>3=Illegal format specified for CPVAL<br>>3=Other equipment -specific error 4-63 Reserved |
| CPNAME  | A                            | Max40 | Command Parameter Name                                                                                                                                                                     |
| CPVAL   | A,B,BOOLEAN,<br>Fx,Ix,J,Ux   |       | Command Parameter value                                                                                                                                                                    |
| DATAID  | U4                           |       | Data ID<br>Integrator normally send messages with DATAID=0                                                                                                                                 |

| Name    | Format                     | Size | Description                                                                                                                                                                                                                                                                                                                                                             |
|---------|----------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DRACK   | B                          | 1    | 0=Accepted<br>1=Denied,Insufficient space<br>2=Denied,Invalid format<br>3=Denied,At least one RPTID already defined<br>4=Denied,At least VID does not exist<br>>4=Other errors 5-63 Reserved                                                                                                                                                                            |
| EAC     | B                          | 1    | 0=Accepted<br>1=Denied,At least one constant does not exist<br>2=Denied,Busy<br>3=Denied,At least one constant out of range<br>>3=Other equipment – specific error 4-63Reserved                                                                                                                                                                                         |
| ECDEF   | A,B,BOOLEAN,<br>Fx,Ix,J,Ux |      | Equipment constant default value                                                                                                                                                                                                                                                                                                                                        |
| ECID    | U4                         |      | Equipment constant ID                                                                                                                                                                                                                                                                                                                                                   |
| ECMAX   | A,B,BOOLEAN,<br>Fx,Ix,J,Ux |      | Equipment constant maximum value                                                                                                                                                                                                                                                                                                                                        |
| ECMIN   | A,B,BOOLEAN,<br>Fx,Ix,J,Ux |      | Equipment constant minimum value                                                                                                                                                                                                                                                                                                                                        |
| ECNAME  | A                          |      | Equipment constant name                                                                                                                                                                                                                                                                                                                                                 |
| ECV     | A,B,BOOLEAN,<br>Fx,Ix,J,Ux |      | Equipment constant value                                                                                                                                                                                                                                                                                                                                                |
| ERACK   | B                          | 1    | 0=Accepted<br>1=Denied,At least one CEID does not exist<br>>1=Other Errors 2-63 Reserved                                                                                                                                                                                                                                                                                |
| HACK    | B                          | 1    | 0=Acknowledge,command has been performed<br>(Stocker will use value of 4 instead of 0)<br><br>1=Command does not exist<br>2=Can not perform now<br>3=At least one parameter is invalid<br>4=Acknowledge,Command will be performed with completion<br>signaled later by an event<br>5=Rejected, Already in desired condition<br>6=No such object exists<br>7-63 Reserved |
| LRACK   | B                          | 1    | 0=Accepted<br>1=Denied,Insufficient space<br>2=Denied,Invalid format<br>3=Denied,At least one CEID link already denied<br>4=Denied,At least one CEID does not exist<br>5=Denied,At least one RPTID does not exist<br>>5= Other errors 6-63 Reserved                                                                                                                     |
| MDLN    | A                          | Max6 | Equipment model type, 6 bytes max                                                                                                                                                                                                                                                                                                                                       |
| MEXP    | A                          |      | SxxFyy                                                                                                                                                                                                                                                                                                                                                                  |
| MHEAD   | B                          |      | SECS message block header associated with message block in<br>error                                                                                                                                                                                                                                                                                                     |
| OBJSPEC | A                          |      | A test string that has an internal format and that is used to point<br>to a specific object instance<br>Integrator always have to set “Null” for OBJSPEC and send it                                                                                                                                                                                                    |
| OFLACK  | B                          |      | 0=Off-line Accepted 1-63Reserved                                                                                                                                                                                                                                                                                                                                        |

| Name    | Format                       | Size | Description                                                                                 |
|---------|------------------------------|------|---------------------------------------------------------------------------------------------|
| ONLACK  | B                            |      | 0=Off-line Accepted<br>1=On-line not allowed<br>2=Equipment already On-line<br>3-63Reserved |
| RCMD    | A                            |      | Remote command code or string                                                               |
| RPTID   | U4                           |      | Report ID                                                                                   |
| SHEAD   | B                            | Max6 | Stored header related to the transaction timer                                              |
| SOFTREV | A                            |      | Software revision code                                                                      |
| SV      | A,B,BOOLEAN,<br>Fx,Ix,J,L,Ux |      | Status variable value                                                                       |
| SVID    | U4                           |      | Status variable ID                                                                          |
| SVNAME  | A                            |      | Status variable name                                                                        |
| TEXT    | A                            |      | Single line of characters                                                                   |
| TIACK   | B                            | 1    | Equipment Acknowledgement code<br>0=Correct 1=Error,Too many SVIDs 2-63 Reserved            |
| TID     | B                            | 1    | 0=Single or main terminal >0=Additional terminals at the same equipment                     |
| TIME    | A                            | 16   | YYMMDDhhmmsscc                                                                              |
| UNITS   | A                            |      | As allowed by SEMI E5 Section 9                                                             |
| V       | A,B,BOOLEAN,<br>Fx,Ix,J,L,Ux |      | Variable data                                                                               |
| VID     | U4                           |      | Variable                                                                                    |

## 4.2 Variable Table

This section explains the variable and VID of stocker.

### 4.2.1 VID Assignment of Variables

The number system of VID that is assigned to the variable of stocker is shown in next table.

**Table 4 VID Assignment of Variables**

| Variable type                    | VID      |
|----------------------------------|----------|
| VID assigned by Selete Spec.     | 1--300   |
| Variable defined by GEM          | 1--50    |
| Variable defined by SEM          | 51--200  |
| Variable defined by Selete Spec. | 201--300 |
| VID assigned by Supplier         | 301--    |

Note: Even if it is the variable defined by GEM or SEM, when VID is not defined by Selete specification but VID is defined uniquely by the supplier, the number after No. 301 is assigned.

## 4.2.2 Variables For Stocker

The variable for stocker is shown below.

Item description of a table

VID : ID of a variable  
 Variable name : The name of a variable  
 Class : The class of a variable data  
 Format : The format of a variable data  
 Description : Description of a variable

**Table 5 Variables used by Stocker, which are defined by GEM**

| VID | Variable Name                  | Class | Format         | Description                                                                                                                                 |
|-----|--------------------------------|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | AlarmID                        | DVVAL | <U4>           | Alarm ID                                                                                                                                    |
| 2   | EstablishCommunicationsTimeout | ECV   | <U2>           | Time Interval for sending a S1F13 message when establishing communication                                                                   |
| 3   | AlarmsEnabled                  | SV    | <L <ALID> ...> | Enable Alarm list                                                                                                                           |
| 4   | AlarmsSet                      | SV    | <L <ALID> ...> | ACTIVE Alarm list                                                                                                                           |
| 5   | Clock                          | SV    | <A[16]>        | Current Time                                                                                                                                |
| 6   | ControlState                   | SV    | B,U1           | 1=Off-line/Equipment Off-line<br>2=Off -line/in the process of Off-line<br>3=Off -line/Host Off-line<br>4=On-line/Local<br>5=On-line/Remote |
| 7   | EventsEnabled                  | SV    | <L <CEID> ...> | Enabled event list                                                                                                                          |

**Table 6 Variables used by Stocker, which are defined by Stocker SEM**

| VID | Variable Name   | Class | Format                         | Description                                                                                             |
|-----|-----------------|-------|--------------------------------|---------------------------------------------------------------------------------------------------------|
| 51  | ActiveCarriers  | SV    | <L <CarrierInfo> ...>          | List current status of all carrier information in the SC Database                                       |
| 52  | ActiveTransfers | SV    | <L <TransferCommand> ...>      | List current status of all active transfer commands                                                     |
| 53  | ActiveZones     | SV    | <L <ZoneData> ...>             | List current status associated with all zones being used by the SC                                      |
| 54  | CarrierID       | DVVAL | <A[1..64]>                     | Unique ID of carrier                                                                                    |
| 55  | CarrierInfo     | DVVAL | <L[2] <CarrierID><CarrierLoc>> | All database information associated with a particular carrier                                           |
| 56  | CarrierLoc      | DVVAL | <A[1..64]>                     | Unique location of the carrier within the stocker as reported by the SC                                 |
| 57  | CommandName     | DVVAL | <A[1..20]>                     | Name of Host issued remote command                                                                      |
| 58  | CommandID       | DVVAL | <A[1..64]>                     | Remote Command ID                                                                                       |
| 59  | CommandInfo     | DVVAL | <L[2] <CommandID><Priority>>   | Command information associated with a particular transfer command                                       |
| 60  | Dest            | DVVAL | <A[1..64]>                     | Destination location identifier                                                                         |
| 61  | EmptyCarrier    | DVVAL | <U2>                           | Flag which denotes whether the carrier is empty or not empty<br>0=Empty 1=Not Empty                     |
| 62  | EqpName         | ECV   | <A[1..32]>                     | Unique ID of the SC                                                                                     |
| 63  | ErrorID         | DVVAL | <A[1..64]>                     | Unique ID of an error                                                                                   |
| 64  | HandoffType     | DVVAL | <U2>                           | Denotes the type of handoff that occurred at the equipment ownership transfer point 1=MANUAL2=AUTOMATED |
| 65  | IDReadStatus    | DVVAL | <U2>                           | Result Code of an ID read event<br>0=Success 1=Failure 2=Duplicate 3=Mismatch                           |
| 66  | PortType        | DVVAL | <A[1..32]>                     | Definition of the type of port associated with the carriers current location OP/BP/LP                   |
| 67  | Priority        | DVVAL | <U2>                           | Remote command priority<br>1--99 Available for use 1:Lowest,99:Highest                                  |
| 68  | RecoveryOptions | DVVAL | <A[1..64]>                     | List of options that the Host may use to try to recover specific stocker error                          |
| 69  | ResultCode      | DVVAL | <U2>                           | Result code of a stocker system command associated with the command completion event                    |
| 70  | Source          | DVVAL | <A[1..64]>                     | Source location unique identifier                                                                       |
| 71  | StockerUnitID   | SV    | <A[1..32]>                     | Unique identification of a stocker unit (Port, Transfer agent)                                          |

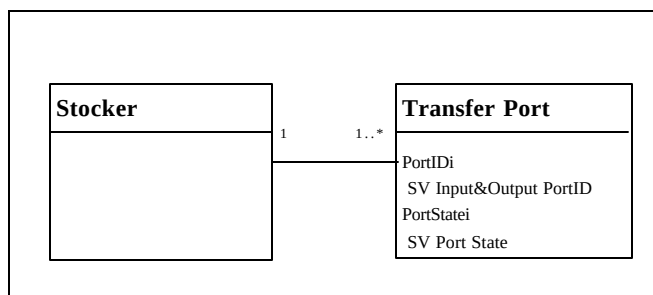
*Stocker Online Requirement Document (Version. 2.31)*

|    |                   |       |                                                                             |                                                                   |
|----|-------------------|-------|-----------------------------------------------------------------------------|-------------------------------------------------------------------|
| 72 | StockerUnitInfo   | DVVAL | <L[2] <StockerDeviceID><br><StockerDeviceState>>                            | Information associated with a particular stocker unit             |
| 73 | StockerUnitState  | DVVAL | <U2>                                                                        | The state of the stocker unit                                     |
| 74 | TransferCommand   | DVVAL | <L[2] <CommandInfo> <TransferInfo>>                                         | Information associated with a particular Transfer command         |
| 75 | TransferInfo      | DVVAL | <L[3] <CarrierID><CarrierLoc><Dest>>                                        | Carrier information associated with a particular Transfer command |
| 76 | ZoneCapacity      | DVVAL | <U2>                                                                        | Available capacity of a particular zone                           |
| 77 | ZoneData          | DVVAL | <L[2] <ZoneName><ZoneCapacity>>                                             | Information associated with a particular zone                     |
| 78 | ZoneName          | DVVAL | <A[1..64]>                                                                  | Alphanumeric name of a particular zone                            |
| 79 | SCState           | SV    | <U2>                                                                        | SCState<br>1=SCInit, 2=Paused, 3=Auto, 4=Pausing                  |
| 80 | PortID            | DVVAL | <A[1..64]>                                                                  | Transfer Port ID                                                  |
| 81 | CurrentPortStates | SV    | <L<br><L[2]<br><A[1..64] PortID><br><U2 PortTransferState><br>><br>...<br>> | The IDs and state of all Ports in a stocker are returned.         |

### 4.2.3 PortTransferState and CurrentPortStates

PortTransferState represents a state that Transfer Port has.

CurrentPortStates is a list of “PortID” and “PortTransferState” in all TransferPorts that a Stocker has.



**Figure 4 Model for Stocker and Transfer Port**

The values that PortTransferState can take are shown in the following table.

**Table 7 Values of PortTransferState**

| Variable          | Value | State                    |                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------|-------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PortTransferState | 2     | Normal<br>InService      | A state in which input/output to and from the shelf can be performed normally.                                                                                                                                                                                                                                                                                                                             |
|                   | 1     | Abnormal<br>OutOfService | <p>An abnormal state</p> <p>Even when a carrier input from or output to the Transfer Port cannot be performed due to irregularities in portions other than the Transfer Port, SC regards the Transfer Port's PortTransferState as irregular. Example: When BP fails at an output port, SC regards the related LP1 and LP2 as out-of-service respectively, and performs an event-reporting. (See below)</p> |

## 5 SECS Messages

This chapter explains the stream and the function of SECS message that are used by the stocker.

### 5.1 Message List

SECS message that is supported by the stocker as a minimum is shown in the next table. A Selete's assuming integrator supports SECS messages that are marked by "X" in the column of "Use" in the table.

Item description of a table

- SECS : The stream and the function of a message.
- Use : The message used by a Selete's assuming integrator is marked with "X."
- Transmission : The message sent to a Stocker from an integrator is marked with "H."  
The message sent to an integrator from a Stocker is marked with "E."
- Format : The message format that is shown in the message definition of SEMI E5.
- Example of data : The example of data of a message is described in SML\* format.

**Table 8 SECS Messages Used by Stocker**

| SECS | Use | Send | Format                                            | Example                                                                  |
|------|-----|------|---------------------------------------------------|--------------------------------------------------------------------------|
| S1F1 | X   | H,E  | S1F1 W<br>. /* Header only */                     | S1F1 W<br>.                                                              |
| S1F2 | X   | H    | S1F2<br><L>.                                      | S1F2<br><L>.                                                             |
|      | X   | E    | S1F2<br><L [2]<br><A MDLN ><br><A SOFTREV ><br>>. | S1F2<br><L [2]<br><A 'AMHS'><br><A 'VER1.0'><br>>.                       |
| S1F3 | X   | H    | S1F3 W<br><L<br><U4 SVID><br>...<br>>.            | S1F3 W<br><L [2]<br><U4 6> /* ControlState*/<br><U4 5> /* Clock */<br>>. |

\*SML (r) is a registered trademark, and SECS Message Language (tm) is a trademark of GW Associates, Inc.

| SECS  | Use | Send | Format                                                                          | Example                                                                                                           |
|-------|-----|------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| S1F4  | X   | E    | S1F4<br><L<br><SV><br>...<br>>.                                                 | S1F4<br><L [2]<br><U1 5> /* ControlState */<br><A[16] '2002011710334600'> /* Clock */<br>>                        |
| S1F11 | -   | H    | S1F11 W<br><L<br><U4 SVID><br>...<br>>.                                         | S1F11 W<br><L [1]<br><U4 6> /* SVID(ControlState) */<br>>.                                                        |
| S1F12 | -   | E    | S1F12<br><L<br><L [3]<br><U4 SVID><br><A SVNAME><br><A UNITS><br>><br>...<br>>. | S1F12<br><L [1]<br><L [3]<br><U4 6> /* SVID */<br><A 'ControlState'> /* SVNAME */<br><A "> /* UNITS */<br>><br>>. |
| S1F13 | X   | H    | S1F13 W<br><L>.                                                                 | S1F13 W<br><L>.                                                                                                   |

| SECS  | Use | Send | Format                                                                    | Example                                                                                                           |
|-------|-----|------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
|       | X   | E    | S1F13 W<br><L [2]<br><A MDLN><br><A SOFTREV><br>>.                        | S1F13 W<br><L [2]<br><A 'AMHS'><br><A 'VER1.0'><br>>.                                                             |
| S1F14 | X   | H    | S1F14<br><L [2]<br><B COMMACK><br><L><br>>.                               | S1F14<br><L [2]<br><B 0x00><br><L><br>>.                                                                          |
|       | X   | E    | S1F14<br><L [2]<br><B COMMACK><br><L [2]<br><A MDLN><br><A SOFTREV><br>>. | S1F14<br><L [2]<br><B 0x00><br><L [2]<br><A 'AMHS'><br><A 'VER1.0'><br>>.                                         |
| S1F15 | X   | H    | S1F15 W<br>. /* Header only */                                            | S1F15 W<br>.                                                                                                      |
| S1F16 | X   | E    | S1F16<br><B OFLACK>.                                                      | S1F16<br><B 0x00>.                                                                                                |
| S1F17 | X   | H    | S1F17 W<br>. /* Header only */                                            | S1F17 W<br>.                                                                                                      |
| S1F18 | X   | E    | S1F18<br><B ONLACK>.                                                      | S1F18<br><B 0x00>.                                                                                                |
| S2F13 | -   | H    | S2F13 W<br><L<br><U4 ECID><br>...<br>>.                                   | S2F13 W<br><L [1]<br><U4 2> /* ECID(Establish Communication Timeout) */<br>>.                                     |
| S2F14 | -   | E    | S2F14<br><L<br><Ux ECV><br>...<br>>.                                      | S2F14<br><L [1]<br><U2 10> /* ECV */<br>>.                                                                        |
| S2F15 | -   | H    | S2F15 W<br><L<br><L [2]<br><U4 ECID><br><Ux ECV><br>><br>...<br>>.        | S2F15 W<br><L [1]<br><L [2]<br><U4 2> /* ECID(Establish Communication Timeout) */<br><U2 20> /* ECV */<br>><br>>. |
| S2F16 | -   | E    | S2F16<br><B EAC>.                                                         | S2F16<br><B 0x00>.                                                                                                |
| S2F17 | X   | HE   | S2F17 W<br>. /* Header only */                                            | S2F17 W<br>.                                                                                                      |
| S2F18 | X   | HE   | S2F18<br><A TIME>.                                                        | S2F18<br><A '1999123123595999'>.                                                                                  |
| S2F29 | -   | H    | S2F29 W<br><L<br><U4 ECID><br>...<br>>.                                   | S2F29 W<br><L [1]<br><U4 2> /* ECID(Establish Communication Timeout) */<br>>.                                     |

| SECS  | Use | Send | Format                                                                                                                    | Example                                                                                                                                                                                                                                                                                                  |
|-------|-----|------|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S2F30 | -   | E    | S2F30<br><L<br><L [6]<br><U4 ECID><br><A ECNAME><br><Ux ECMIN><br><Ux ECMAx><br><Ux ECDEF><br><A UNITS><br>><br>...<br>>. | S2F30<br><L [1]<br><L [6]<br><U4 2> /* ECID */<br><A 'Establish Communications Timeout'><br>/* ECNAME */<br><U2 0> /* ECMIN */<br><U2 XXX> /* ECMAx */<br><U2 10> /* ECDEF */<br><A "> /* UNITS */<br>>.                                                                                                 |
| S2F31 | -   | H    | S2F31 W<br><A TIME>.<                                                                                                     | S2F31 W<br><A '1999123123595999'>.<                                                                                                                                                                                                                                                                      |
| S2F32 | -   | E    | S2F32<br><B TIACK>.<                                                                                                      | S2F32<br><B 0x00>.<                                                                                                                                                                                                                                                                                      |
| S2F33 | X   | H    | S2F33 W<br><L [2]<br><U4 DATAID><br><L<br><L [2]<br><U4 RPTID><br><L<br><U4 VID><br>...<br>><br>><br>...<br>><br>>.       | S2F33 W<br><L [2]<br><U4 0> /* DATAID */<br><L [2] /* Two kinds of report format */<br><L [2]<br><U4 1> /* RPTID */<br><L [1]<br><U4 62> /* VID(EqpName) */<br>><br>><br><L [2]<br><U4 2> /* RPTID */<br><L [2]<br><U4 62> /* VID(EqpName) */<br><U4 74> /* VID(TransferCommand) */<br>><br>><br>><br>>. |
| S2F34 | X   | E    | S2F34<br><B DRACK>.<                                                                                                      | S2F34<br><B 0x00>.<                                                                                                                                                                                                                                                                                      |
| S2F35 | X   | H    | S2F35 W<br><L [2]<br><U4 DATAID><br><L<br><L [2]<br><U4 CEID><br><L<br><U4 RPTID><br>...<br>><br>><br>...<br>><br>>.      | S2F35 W<br><L [2]<br><U4 0> /* DATAID */<br><L [2] /* Two types of event */<br><L [2]<br><U4 3> /* CEID(Control State Remote) */<br><L [1]<br><U4 1> /* RPTID */<br>><br>><br><L [2]<br><U4 108> /* CEID(TransferInitiated) */<br><L [2]<br><U4 1> /* RPTID */<br>><br>><br>><br>>.                      |

| SECS  | Use | Send | Format                                                                                          | Example                                                                                                                                                                                                                                                |
|-------|-----|------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S2F36 | X   | E    | S2F36<br><B LRACK>.                                                                             | S2F36<br><B 0x00>.                                                                                                                                                                                                                                     |
| S2F37 | X   | H    | S2F37 W<br><L [2]<br><BOOLEAN CEED><br><L<br><U4 CEID><br>...<br>><br>>.                        | S2F37 W<br><L [2]<br><BOOLEAN T> /* CEED */<br><L [2] /* Two types of event */<br><U2 4> /* CEID(Control State Remote) */<br><U4 108> /* CEID(TransferInitiated) */<br>><br>>.                                                                         |
| S2F38 | X   | E    | S2F38<br><B ERACK>.                                                                             | S2F38<br><B 0x00>.                                                                                                                                                                                                                                     |
| S2F41 | X   | H    | S2F41 W<br><L [2]<br><A RCMD><br><L<br><L [2]<br><A CPNAME><br><A CPVAL><br>><br>...<br>><br>>. | S2F41 W<br><L [2]<br><A 'CANCEL'> /* RCMD */<br><L [1]<br><L [2]<br><A 'COMMANDID'> /* CPNAME */<br><A '1999110112000000'> /* CPVAL */<br>><br>><br>>.                                                                                                 |
| S2F42 | X   | E    | S2F42<br><L [2]<br><B HBACK><br><L<br><L [2]<br><A CPNAME><br><B CPACK><br>><br>...<br>><br>>.  | /*No Error exist<br>S2F42<br><L [2]<br><B 0x00> /* HBACK */<br><L [0]<br>><br>>.<br><br>/*Invalid parameter exist<br>S2F42<br><L [2]<br><B 0x03> /* HBACK */<br><L [1]<br><L [2]<br><A 'CPNAME1'> /* CPNAME */<br><B 0x01> /* CPACK */<br>><br>><br>>. |



| SECS | Use | Send | Format                                                                         | Example                                                                                                                                                                                                      |
|------|-----|------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      |     |      |                                                                                | >.                                                                                                                                                                                                           |
| S5F1 | X   | E    | S5F1 W<br><L [3]<br><B ALCD><br><U4 ALID><br><A ALTX><br>>.                    | S5F1 W<br><L [3]<br><B 0x80> /* ALCD */<br><U4 1> /* ALID */<br><A 'ALARM TEXT'> /* ALTX */<br>>.                                                                                                            |
| S5F2 | X   | H    | S5F2<br><B ACKC5>.                                                             | S5F2<br><B 0x00>.                                                                                                                                                                                            |
| S5F3 | X   | H    | S5F3 W<br><L [2]<br><B ALED><br><U4 ALID><br>>.                                | S5F3 W<br><L [2]<br><B 0x80> /* ALED */<br><U4 1> /* ALID */<br>>.                                                                                                                                           |
| S5F4 | X   | E    | S5F4<br><B ACKC5>.                                                             | S5F4<br><B 0x00>.                                                                                                                                                                                            |
| S5F5 | -   | H    | S5F5 W<br><U4 ALID ...>.                                                       | S5F5 W<br><U4 [2] 1 2>. /* Two ALIDs */                                                                                                                                                                      |
| S5F6 | -   | E    | S1F12 W<br><L<br><L [3]<br><B ALCD><br><U4 ALID><br><A ALTX><br>><br>...<br>>. | S1F12 W<br><L [2]<br><L [3]<br><B 0x80> /* ALCD */<br><U4 1> /* ALID */<br><A 'ALARM TEXT'> /* ALTX */<br>><br><L [3]<br><B 0x00> /* ALCD */<br><U4 2> /* ALID */<br><A 'ALARM TEXT2'> /* ALTX */<br>><br>>. |

| SECS  | Use | Send | Format                                                                                                                      | Example                                                                                                                                                                                        |
|-------|-----|------|-----------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S6F11 | X   | E    | S6F11 W<br><L [3]<br><U4 DATAID><br><U4 CEID><br><L<br><L [2]<br><U4 RPTID><br><L<br><V><br>...<br>><br>><br>...<br>><br>>. | S6F11 W<br><L [3]<br><U4 0> /* DATAID */<br><U4 3> /* CEID(Control State Remote) */<br><L [1]<br><L [2]<br><U4 1> /* RPTID */<br><L [1]<br><A 'AMHS001'> /* V(EqpName) */<br>><br>><br>><br>>. |
| S6F12 | X   | H    | S6F12<br><B ACKC6>.                                                                                                         | S6F12<br><B 0x00>.                                                                                                                                                                             |
| S9F1  | X   | E    | S9F1<br><MHEAD>.                                                                                                            | S9F1<br><B 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00>.                                                                                                                                 |
| S9F3  | X   | E    | S9F3<br><MHEAD>.                                                                                                            | S9F3<br><B 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00>.                                                                                                                                 |
| S9F5  | X   | E    | S9F5<br><MHEAD>.                                                                                                            | S9F5<br><B 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00>.                                                                                                                                 |
| S9F7  | X   | E    | S9F7<br><MHEAD>.                                                                                                            | S9F7<br><B 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00>.                                                                                                                                 |
| S9F9  | X   | E    | S9F9<br><SHEAD>.                                                                                                            | S9F9<br><B 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00>.                                                                                                                                 |
| S9F11 | X   | E    | S9F11<br><MHEAD>.                                                                                                           | S9F11<br><B 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00>.                                                                                                                                |
| S10F3 | X   | H    | S10F3 W<br><L [2]<br><B TID><br><A TEXT><br>>.                                                                              | S10F3 W<br><L [2]<br><B 0x00> /* TID */<br><A 'SCREEN TEXT'> /* TEXT */<br>>.                                                                                                                  |
| S10F4 | X   | E    | S10F4<br><B ACKC10>.                                                                                                        | S10F4<br><B 0x00>.                                                                                                                                                                             |

## 6 Remote Commands

This chapter explains the remote command that is supported by the stocker.

### 6.1 ABORT

| CPNAME    | CPVAL    |                                                    |
|-----------|----------|----------------------------------------------------|
|           | Format   | Comment                                            |
| COMMANDID | A[1..64] | Command parameter name for transfer remote command |

SECS communication format

```

S2F41 W
<L [2]
  <A 'ABORT'> /* RCMD */
  <L [1]
    <L [2]
      <A 'COMMANDID'> /* CPNAME */
      <A CommandID> /* CPVAL */
    >
  >
>.

S2F42
<L [2]
  <B 0x00> /* HCACK */
  <L [0]
  >
>.
```

---

## 6.2 CANCEL

| CPNAME    | CPVAL    |                                                    |
|-----------|----------|----------------------------------------------------|
|           | Format   | Comment                                            |
| COMMANDID | A[1..64] | Command parameter name for transfer remote command |

### SECS Communication format

---

```

S2F41 W
<L [2]
  <A 'CANCEL'> /* RCMD */
  <L [1]
    <L [2]
      <A 'COMMANDID'> /* CPNAME */
      <A CommandID> /* CPVAL */
    >
  >
>.
```

```

S2F42
<L [2]
  <B 0x00> /* HCACK */
  <L [0]
  > >.
```

---

### 6.3 INSTALL

| CPNAME     | CPVAL    |                                                                     |
|------------|----------|---------------------------------------------------------------------|
|            | Format   | Comment                                                             |
| CARRIERID  | A[1..64] | Carrier ID for registering or alternating Carrier information       |
| CARRIERLOC | A[1..64] | Carrier location for registering or alternating Carrier information |

#### SECS Communication format

---

```

S2F41 W
<L [2]
  <A 'INSTALL'> /* RCMD */
  <L [2]
    <L [2]
      <A 'CARRIERID'> /* CPNAME */
      <A CarrierID> /* CPVAL */
    >
    <L [2]
      <A 'CARRIERLOC'> /* CPNAME */
      <A CarrierLoc> /* CPVAL */
    >
  >
>
>.
```

```

S2F42
<L [2]
  <B 0x00> /* HCAACK */
  <L [0]
  >
>.
```

---

## 6.4 LOCATE

| CPNAME    | CPVAL    |                                                                   |
|-----------|----------|-------------------------------------------------------------------|
|           | Format   | Comment                                                           |
| CARRIERID | A[1..64] | Set Carrier ID that you would like to know it's current location. |

### SECS Communication format

---

```

S2F41 W
<L [2]
  <A 'LOCATE'> /* RCMD */
  <L [1]
    <L [2]
      <A 'CARRIERID'> /* CPNAME */
      <A CarrierID> /* CPVAL */
    >
  >
>.

S2F42
<L [3]
  <B 0x00> /* HBACK */
  <L [0]
  >
>.
```

---

## 6.5 PAUSE

| CPNAME | CPVAL  |         |
|--------|--------|---------|
|        | Format | Comment |
| None   |        |         |

### SECS Communication format

```

S2F41 W
<L [2]
  <A 'PAUSE'> /* RCMD */
  <L [0]
  >
>.

```

```

S2F42
<L [2]
  <B 0x00> /* HBACK */
  <L [0]
  >
>.

```

## 6.6 REMOVE

| CPNAME    | CPVAL    |         |
|-----------|----------|---------|
|           | Format   | Comment |
| CARRIERID | A[1..64] |         |

### SECS Communication format

```

S2F41 W
<L [2]
  <A 'REMOVE'> /* RCMD */
  <L [1]
    <L [2]
      <A 'CARRIERID'> /* CPNAME */
      <A 'Carrier1'> /* CPVAL */
    >
  >
>.

```

```

S2F42
<L [2]
  <B 0x00> /* HBACK */
  <L [0]
  >
>.

```

## 6.7 RESUME

| CPNAME | CPVAL  |         |
|--------|--------|---------|
|        | Format | Comment |
| None   |        |         |

### SECS Communication format

---

```

S2F41 W
<L [2]
  <A 'RESUME'> /* RCMD */
  <L [0]
  >
>.
```

```

S2F42
<L [2]
  <B 0x00> /* HCACK */
  <L [0]
  >
>.
```

---

## 6.8 TRANSFER

| CPNAME       | CPVAL    |                                                                                                                                                   |
|--------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Format   | Comment                                                                                                                                           |
| COMMANDINFO  | L[2]     |                                                                                                                                                   |
| COMMANDID    | A[1..64] |                                                                                                                                                   |
| PRIORITY     | U2       | 0=Not valid<br>1=Lowest 99=Highest                                                                                                                |
| TRANSFERINFO | L[3]     |                                                                                                                                                   |
| CARRIERID    | A[1..64] |                                                                                                                                                   |
| SOURCE       | A[1..64] | Transfer Source(can be omitted) <ul style="list-style-type: none"> <li>• input port</li> <li>• ZoneName</li> <li>• Location of Storage</li> </ul> |
| DEST         | A[1..64] | Destination <ul style="list-style-type: none"> <li>• Loading port(LP)</li> <li>• ZoneName</li> </ul>                                              |

### SECS Communication format

```

S2F49 W
<L [4]
  <U4 0> /* DATAID */
  <A "> /* OBJSPEC */
  <A 'TRANSFER'> /* RCMD */
  <L [2]
    <L [2]
      <A 'COMMANDINFO'>
      <L [2]
        <L [2]
          <A 'COMMANDID'>
          <A CommandID>
        >
      <L [2]
        <A 'PRIORITY'>
        <U2 Priority>
      >
    >
  >
  <L [2]
    <A 'TRANSFERINFO'>
    <L [3]
      <L [2]
        <A 'CARRIERID'>
        <A CarrierID>
      >
      <L [2]
        <A 'SOURCE'>
        <A Source>
      >
      <L [2]
        <A 'DEST'>
    >
  >

```

```

                                <A Dest>
                                >
                                >
                                >
                                >
                                >.

S2F50
<L [2]
    <B 0x00> /* HCKACK */
    <L [0]
    >
    >.

```

---

## 7 Collection Events

The stocker supports the event report that is based on "Dynamic Event Report Configuration" described in GEM. This chapter explains the event that is used by the stocker.

### 7.1 CEID Assignment of Collection Events

The number system of CEID that is assigned to the event of stocker is shown in next table.

**Table 9 CEID Assignment of Collection Events**

| Event Type                    | CEID     |
|-------------------------------|----------|
| CEID assigned by Selete Spec. | 1--600   |
| Event defined by GEM          | 1--50    |
| Event defined by SEM          | 51--500  |
| Event defined by Selete Spec. | 501--600 |
| CEID assigned by Supplier     | 601--    |

Note: Even if it is the event defined by GEM or SEM, when CEID is not defined by Selete specification but CEID is defined uniquely by the supplier, then the number after No. 601 is assigned.

### 7.2 Collection Events for Stocker

This section explains the classification of event that is used by the stocker.

#### 7.2.1 Control State Transition Events

| CEID | Event Name           | Previous State | New State         | Report ID |
|------|----------------------|----------------|-------------------|-----------|
| 1    | Equipment Off-line   | ONLINE         | Equipment OFFLINE | 1         |
| 2    | Control State local  | OFFLINE        | ONLINE-LOCAL      | 1         |
|      |                      | ONLINE-REMOTE  |                   |           |
| 3    | Control State Remote | OFFLINE        | ONLINE-REMOTE     | 1         |
|      |                      | ONLINE-LOCAL   |                   |           |

### 7.2.2 SC State Transtion Events

| CEI<br>D | Event Name                                                                | Previous State | New State | Report ID |
|----------|---------------------------------------------------------------------------|----------------|-----------|-----------|
| 51       | AlarmCleared                                                              | ARARMS         | NO ALARMS | 1         |
|          | This specification does not require implementation of AlarmCleared event. |                |           |           |
| 52       | AlarmSet                                                                  | NO ARARMS      | ALARMS    | 1         |
|          | This specification does not require implementation of AlarmSet event.     |                |           |           |
| 53       | SCAutoCompleted                                                           | PAUSED         | AUTO      | 1         |
|          |                                                                           | PAUSING        |           |           |
| 54       | SCAutoInitiated                                                           | None           | SC INIT   | 1         |
| 55       | SCPauseCompleted                                                          | PAUSING        | PAUSED    | 1         |
| 56       | SCPausd                                                                   | SC INIT        | PAUSED    | 1         |
| 57       | SCPauseInitiated                                                          | AUTO           | PAUSING   | 1         |

### 7.2.3 Transfer Command State Transition Events

| CEI<br>D | Event Name              | Previous State | New State    | Report ID |
|----------|-------------------------|----------------|--------------|-----------|
| 101      | TransferAbortCompleted  | ABORTING       | None         | 3         |
| 102      | TransferAbortFailed     | ABORTING       | ACTIVE       | 4         |
| 103      | TransferAbortInitiated  | ACTIVE         | ABORTING     | 4         |
| 104      | TransferCancelCompleted | CANCELING      | None         | 3         |
| 105      | TransferCancelFailed    | CANCELING      | QUEUED       | 4         |
| 106      | TransferCancelInitiated | QUEUED         | CANCELING    | 4         |
| 107      | TransferCompleted       | ACTIVE         | None         | 2         |
| 108      | TransferInitiated       | QUEUED         | TRANSFERRING | 4         |
| 109      | TransferPaused          | TRANSFERRING   | PAUSED       | 2         |
| 110      | TransferResumed         | PAUSED         | TRANSFERRING | 4         |

### 7.2.4 Carrier State Transition Events

| CEI<br>D | Event Name              | Previous State       | New State    | Report ID |
|----------|-------------------------|----------------------|--------------|-----------|
| 151      | CarrierInstallCompleted | None                 | COMPLETED    | 5         |
| 152      | CarrierRemoveCompleted  | INSTALLED            | None         | 5         |
| 153      | CarrierRemoved          | WAIT OUT             | None         | 6         |
| 154      | CarrierResumed          | ALTERNATE            | TRANSFERRING | 7         |
| 155      | CarrierStored           | TRANSFERRING         | COMPLETED    | 5         |
| 156      | CarrierStoredAlt        | TRANSFERRING         | ALTERNATE    | 7         |
| 157      | CarrierTransferring     | WAIT IN<br>COMPLETED | TRANSFERRING | 8         |
| 158      | CarrierWaitIn           | None                 | WAIT IN      | 5         |
| 159      | CarrierWaitOut          | TRANSFERRING         | WAIT OUT     | 9         |
|          |                         | WAIT OUT             |              |           |
| 160      | ZoneCapacityChange      |                      |              | 10        |

### 7.2.5 Crane State Transition events

| CEID | Event Name  | Previous State | New State | Report ID |
|------|-------------|----------------|-----------|-----------|
| 201  | CraneActive | IDLE           | ACTIVE    | 11        |
| 202  | CraneIdle   | ACTIVE         | IDLE      | 11        |

### 7.2.6 Other Events

| CEID | Event Name             | Previous State | New State | Report ID |
|------|------------------------|----------------|-----------|-----------|
| 251  | CarrierIDRead          | None           | None      | 12        |
| 252  | CarrierLocateCompleted | None           | None      | 13        |
| 253  | IDReaderError          | None           | None      | 12        |

### 7.2.7 Port State Transition events

| CEID | Event Name                                                                                                               | Previous State       | New State    | Report ID |
|------|--------------------------------------------------------------------------------------------------------------------------|----------------------|--------------|-----------|
| 301  | PortInService                                                                                                            | OutOfService<br>None | InService    | 14        |
|      | This event will be reported by SC when the state of the PortTransferService mentioned above has changed to InService.    |                      |              |           |
| 302  | PortOutOfService                                                                                                         | InService<br>None    | OutOfService | 14        |
|      | This event will be reported by SC when the state of the PortTransferService mentioned above has changed to OutOfService. |                      |              |           |

### 7.3 Event Reports for Stocker

The next table shows, as an example of a dynamic event report, a report format that is used by an Integrator for the stocker.

| Report ID | Variable                                               | Report Format                                                                                                                                                                                                                                        |
|-----------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1         | EqpName                                                | <L[1]<br><A EqpName><br>>                                                                                                                                                                                                                            |
| 2         | EqpName<br>TransferCommand<br>CarrierLoc<br>ResultCode | <L[4]<br><A EqpName><br><L [2] /*TransferCommand>*/<br><L[2] /* CommandInfo */<br><A CommandID><br><U2 Priority><br>><br><L[3] /* TransferInfo */<br><A CarrierID><br><A CarrierLoc><br><A Dest><br>><br>><br><A CarrierLoc><br><U2 ResultCode><br>> |
| 3         | EqpName<br>TransferCommand<br>CarrierLoc               | <L[3]<br><A EqpName><br><L[2] /*TransferCommand>*/<br><L[2] /* CommandInfo */<br><A CommandID><br><U2 Priority><br>><br><L[3] /* TransferInfo */<br><A CarrierID><br><A CarrierLoc><br><A Dest><br>><br>><br>< A CarrierLoc ><br>>                   |
| 4         | EqpName<br>TransferCommand                             | <L[2]<br><A EqpName><br><L[2] /*TransferCommand>*/<br><L[2] /* CommandInfo */<br><A CommandID><br><U2 Priority><br>><br><L[3] /* TransferInfo */<br><A CarrierID><br><A CarrierLoc><br><A Dest><br>><br>><br>>                                       |

| Report ID | Variable                                           | Report Format                                                                                 |
|-----------|----------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 5         | EqpName<br>CarrierID<br>CarrierLoc                 | <L[3]<br><A EqpName><br><A CarrierID><br><A CarrierLoc><br>>                                  |
| 6         | EqpName<br>CarrierID<br>HandoffType                | <L[3]<br><A EqpName><br><A CarrierID><br><U2 HandoffType><br>>                                |
| 7         | EqpName<br>CarrierID<br>CarrierLoc<br>Dest         | <L[4]<br><A EqpName><br><A CarrierID><br><A CarrierLoc><br><A DEST><br>>                      |
| 8         | EqpName<br>CarrierID                               | <L[2]<br><A EqpName><br><A CarrierID><br>>                                                    |
| 9         | EqpName<br>CarrierID<br>CarrierLoc<br>PortType     | <L[4]<br><A EqpName><br><A CarrierID><br><A CarrierLoc><br><A PortType><br>>                  |
| 10        | EqpName<br>ZoneData<br>CarrierID                   | <L[3]<br><A EqpName><br><L[2]<br><A ZoneName><br><U2 ZoneCapacity><br>><br><A CarrierID><br>> |
| 11        | EqpName<br>CommandID                               | <L[2]<br><A EqpName><br><A CommandID><br>>                                                    |
| 12        | EqpName<br>CarrierID<br>CarrierLoc<br>IDReadStatus | <L[4]<br><A EqpName><br><A CarrierID><br><A CarrierLoc><br><U2 IDReadStatus><br>>             |
| 13        | EqpName<br>CarrierID<br>CarrierLoc                 | <L[3]<br><A EqpName><br><A CarrierID><br><A CarrierLoc><br>>                                  |
| 14        | EqpName<br>PortID                                  | <L[2]<br><A EqpName><br><A PortID><br>>                                                       |

## **7.4 Supplements**

### **7.4.1 CarrierWaitIn Event**

1. As CarrierID to be reported in CarrierWaitIn event, SC should report the following ID: either the ID that was acquired by IDReader, or the temporary ID that was generated by SC when abnormalities were detected by CarrierIDRead. (CarrierID should not be blank.)
2. Before CarrierWaitIn event, SC should generate CarrierIDRead event. (The operation of stocker after CarrierWaitIn will be as follows. If a carrier ID read is successful, it will follow a host's Transfer command. If a carrier ID read fails, it will not wait for a host's Transfer command, but will turn into the original operation [for example, an abnormal outgoing process operation] of stocker. For this reason, a host sees CarrierIDRead result and judges whether or not Transfer command is issued.

### **7.4.2 ZoneCapacityChange Event**

1. A host does not positively use the Zone function of Stocker SEM. For this reason, a user should establish only one Zone by a stocker, and let the region be the entire rack. Moreover, Zone should not be set to a port.

## 8 Carrier Location Information

### 8.1 Carrier Location Information in SECS Messages

#### 8.1.1 Notation of Carrier Location in SECS Messages

In case the location information that is used by the message is explained, the name of the location information shown below is used.

The location information showing a unit

| Equipment Type | The name of location information |
|----------------|----------------------------------|
| Stocker        | Stocker ID                       |

The location information showing the unique location (port) in equipment

| Equipment Type | Port Type                                                | The name of location information |
|----------------|----------------------------------------------------------|----------------------------------|
| Stocker        | Input shuttle                                            | Input port                       |
|                | Port, where Crane place a carrier                        | Output port                      |
|                | Area, where people or transfer equipment pick up carrier | Loading port                     |
|                | Buffer port for carrier out -puts.                       | Buffer port                      |
|                | Each port of a shelf                                     | Location in a Storage            |

The location information showing a port group

| Equipment Type | Port Type                     | The name of location information |
|----------------|-------------------------------|----------------------------------|
| Stocker        | Plural (all) ports of a shelf | ZoneName                         |

### 8.1.2 Carrier Location in SECS Message

The location data and the carrier location information that are used by the stocker are shown in next table.

| Name of command or event | Data name of location | Location data of a message          |                 |                                            |
|--------------------------|-----------------------|-------------------------------------|-----------------|--------------------------------------------|
|                          |                       | Storage                             | Input Port      | Output Port                                |
| TRANSFER Command         | SOURCE                | Blank (omitted)                     | Blank (omitted) |                                            |
|                          | DEST                  | ZoneName                            |                 | loading port                               |
| CarrierIDRead            | CarrierLoc            |                                     | input port      |                                            |
| CarrierInstallCompleted  | CarrierLoc            | Shelf Location                      |                 |                                            |
| CarrierLocateCompleted   | CarrierLoc            | The location information at present |                 |                                            |
| CarrierRemoveCompleted   | CarrierLoc            | The location information at present |                 |                                            |
| CarrierResumed           | CarrierLoc            | Shelf Location                      |                 |                                            |
|                          | DEST                  | The same data as TRANSFER command   |                 |                                            |
| CarrierStored            | CarrierLoc            | Shelf Location                      |                 |                                            |
| CarrierStoredAlt         | CarrierLoc            | Shelf Location                      |                 |                                            |
|                          | DEST                  | The same data as TRANSFER command   |                 |                                            |
| CarrierWaitIn            | CarrierLoc            |                                     | input port      |                                            |
| CarrierWaitOut           | CarrierLoc            |                                     |                 | output port<br>buffer port<br>loading port |
| IDReadError              | CarrierLoc            | The location information at present |                 |                                            |
| TransferAbortCompleted   | TransferInfo          | The same data as TRANSFER command   |                 |                                            |
|                          | CarrierLoc            | The location information at present |                 |                                            |
| TransferAbortFailed      | TransferInfo          | The same data as TRANSFER command   |                 |                                            |
| TransferAbortInitiated   | TransferInfo          | The same data as TRANSFER command   |                 |                                            |
| TransferCancelCompleted  | TransferInfo          | The same data as TRANSFER command   |                 |                                            |
|                          | CarrierLoc            | Shelf Location                      | input port      |                                            |

|                         |              |                                     |  |                                            |
|-------------------------|--------------|-------------------------------------|--|--------------------------------------------|
| TransferCancelFailed    | TransferInfo | The same data as TRANSFER command   |  |                                            |
| TransferCancelInitiated | TransferInfo | The same data as TRANSFER command   |  |                                            |
| TransferCompleted       | TransferInfo | The same data as TRANSFER command   |  |                                            |
|                         | CarrierLoc   | Shelf Location                      |  | output port<br>buffer port<br>loading port |
| TransferInitiated       | TransferInfo | The same data as TRANSFER command   |  |                                            |
| TransferPaused          | TransferInfo | The same data as TRANSFER command   |  |                                            |
|                         | CarrierLoc   | The location information at present |  |                                            |
| TransferResumed         | TransferInfo | The same data as TRANSFER command   |  |                                            |
| ZoneCapacityChanged     | ZoneName     | ZoneName                            |  |                                            |

## 9 Interpretation of SEMI Standards

This chapter explains the matters that clarify interpretation out of SEMI standard.

### 9.1 Control State Model for Stocker

The function about the on-line/off-line of a general equipment system is specified in GEM as a control state model. Moreover, the state model of stocker is specified in Stocker SEM as a SC State model. These two state models are used as follows.

The state model of stocker is shown in the following figure.

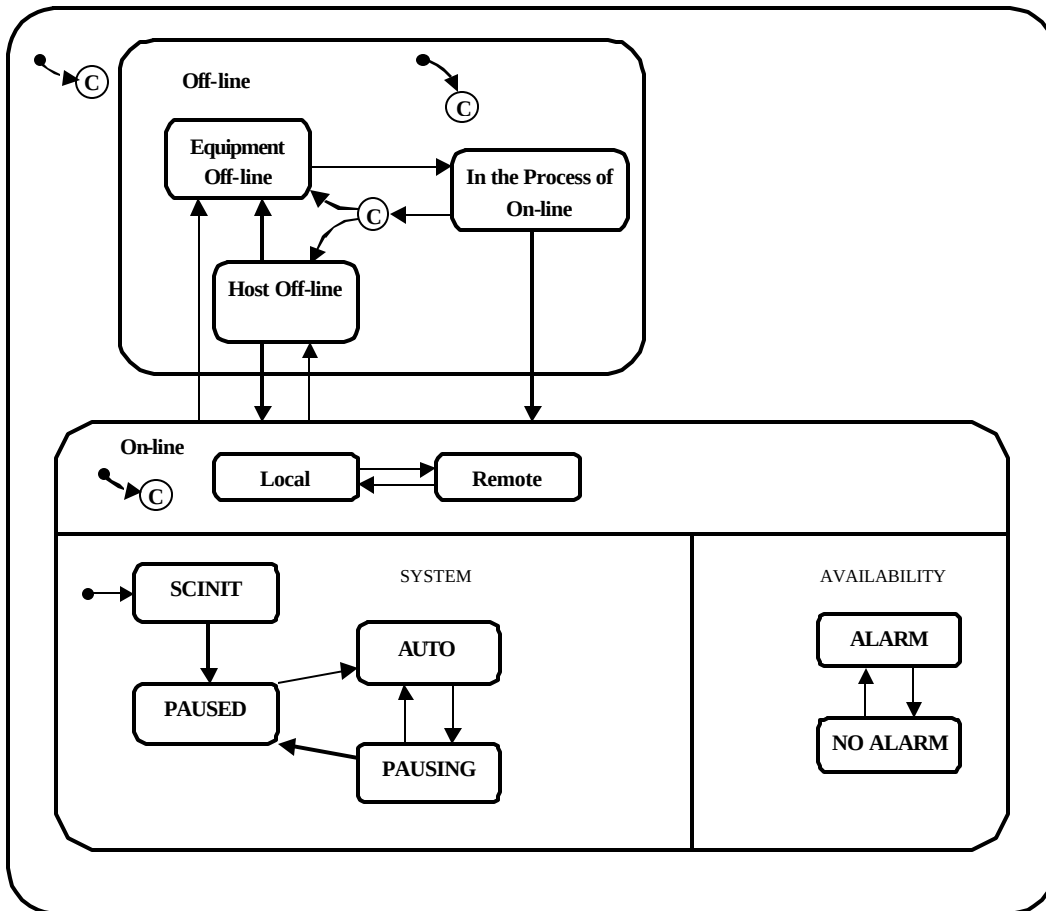
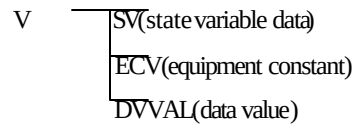


Figure 5 Control State Model of Stocker

## 9.2 VID, SVID and ECID

Variable data (V) show 3 kinds of data as follows.



The relation of VID, SVID, and ECID is as follows.

- Among VIDs, VID that points out SV is called SVID.
- Among VIDs, VID that points out ECV is called ECID.

## 10 Distinction from SEMI Standards.

This section represents the distinction of this Document from StockerSEM.

1. Terminology is added. (Refer to 2. Terminology)
2. Both active and passive modes are required for HSMS Connect Mode. (Refer to 3.1.1. HSMS Connect Mode)
3. Formats of Ux of Data Items are altered to U4. (Refer to 4.1. SECS Data Items)
4. VIDs are specified. (Refer to 4.2. Variable Table)
5. CEIDs are specified. (Refer to 7.2. Collection Events for Stocker)
6. Formats of EventReports are specified by Selete Specification. (Refer to 7.2. Collection Event for Stocker, 6.3. Event Report for Stocker) However, it does not deny the function of "Dynamic Event Report Configuration" described in GEM. Furthermore, if GEM has its function, automatically it is satisfied with this specification. It explains how Host uses event reports.
7. Additional specification to CarrierWaitIn Event. (Refer to 7.4.1. CarrierWaitIn Event)
8. Use of Zone function is not positively recommended. (Refer to 7.4.2. ZoneCapacityChange event)
9. Carrier Location Information is added. (Refer to 8. CarrierLocationInformation)
10. Interpretations of SEMI Standard are clarified. (Refer to 9. Interpretation of SEMI Standards)

## 11 Scenario

For the communication scenario of stocker, refer to the following chapters of AMHS communication scenario specification.

Chapter 2 Common scenario

Chapter 3 Stocker scenario

## 12 Revision History

| Version | Date      | Updating part                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.5     | 99/12/08  | New                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 1.0     | 99/12/27  | <p>The following chapters are added.</p> <p>2.3. Subsidiary Condition</p> <p>7. Interpretation of SEMI standard</p> <p>The followings are updated.</p> <p>3.2.2. Variable that is used by stocker: Control State, EmptyCarrier, and HandoffType,</p> <p>6.2.3. Transport Command StateTransition Event: TransferAbortCompleted (Revision), TransferCancelCompleted (Revision), and TransferCancelFailed (Addition)</p> <p>6.2.7. Event of Selete Original Specification Definition</p> <p>8.1.13. PAUSE and RESUME 8.1.14 Re-connection from Communication Failure</p> <p>8.2.5. Shelf =&gt;Outgoing process multiport BP transfer completed ID Reader is present.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 1.1     | 2000/2/25 | <p>The following chapter is deleted.</p> <p>2.3 subsidiary conditions</p> <p>The following chapters are added.</p> <p>2. Terminology</p> <p>74 Supplement</p> <p>8 Location Information</p> <p>The followings are updated.</p> <p>3.2.1 Description about State about Alarm is deleted.</p> <p>3.2.2 Description about State about Alarm is deleted.</p> <p>3.2.2 Figure of "Entity That Owns State Model" is altered.</p> <p>3.2.2 Explanation of "Service State" and "Port State" is added.</p> <p>4.1 Explanation of "DATAID", "HCACK", and "OBJESPEC" of SECS Data Item is altered.</p> <p>4.1 DATALENGTH, GRANT, and GRANT6 of SECS Data Item are deleted.</p> <p>4.2.2 Explanation of Priority in Table is altered</p> <p>4.2.2 Explanation of IDReadState in Table is deleted</p> <p>4.2.2 Table of Selete Specification Original Definition is altered.</p> <p>4.2.3 Explanation of Selete Specification Original Definition Variable is altered</p> <p>5.1 S2F39/F40 and S6F5/F6 are deleted from Message List.</p> <p>5.1 Data Examples of S2F42 and S2F50 are altered.</p> <p>6.1, 6.2, 6.3, 6.4, 6.6 S2F42 format examples are altered.</p> <p>6.4 ZoneName and CarrierLoc of Locate Command variables are deleted.</p> <p>6.8 S2F50 format example is altered.</p> <p>7.2 Report ID of Event is revised</p> <p>7.2.7 Classification of Event is deleted/added.</p> <p>7.3 Report of Event is added/altered.</p> <p>8.1.2 Table of Location Information for Every Message is altered.</p> |

|            |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                   | <p>9.1 Figure of Event Report of Outgoing Process is added.</p> <p>10 Contents of Scenario is shifted to AMHS Communication Scenario Specification.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>1.2</b> | <b>2000/5/23</b>  | <p>The followings are updated.</p> <p>4.1 Explanation of Format of "Item Description in Table" is altered.</p> <p>4.2.2 Table. class name of (VID=71) is revised</p> <p>4.2.2 Table of "Variable That Is Used by Stocker and That Is Uniquely defined by Selete Specification " has two variables added.</p> <p>4.2.3 Figure of "Stocker and Transfer Port Model". Variables of PortID and PortState have a suffix (i) added.</p> <p>4.2.3.2 Title and Explanation. are altered.</p> <p>5.1 Transmission Directions of S9F1, S9F3, S9F5, S9F7, S9F9, and S9F11 are revised..</p> <p>7.1 Note of Figure "Event CEID System" is revised.</p> <p>7.2.1 New State of (CEID=3) of Table is revised</p> <p>7.2.2 Event Name of (CEID=53, 54, 55, 56, 57).of Table is revised.</p> <p>7.2.4 Event Name and New State of (CEID=151) of Table are revised</p> <p>7.2.4 Old State of (CEID=157) of Table is revised</p> <p>7.2.6 Event Name of (CEID=252) of Table is revised.</p> <p>7.3 Report Format of (Report ID=9, 13) of Table is revised.</p> <p>8.1.2 Table is sorted by Event Name.</p> <p>8.1.2 CarrierLocateCompleted Event is added.</p> |
| <b>1.3</b> | <b>2000/10/23</b> | <p>The followings are updated.</p> <p>3.2.2 SCState and ControlState in the Figure6 are altered to be independent.</p> <p>4.1 ECNAME is added into the SECS data items.</p> <p>4.2.2 The explanation between 64 and 67 for the VID6 in figure5 are deleted.</p> <p>VID79 SCStaet is added into Figure6.</p> <p>9.2 The sentence of "SCSate model replaces the "On-line remote" of the control state model of GEM " is deleted.</p> <p>Figure6 are altered.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>2.0</b> | <b>2000/10/31</b> | <p>The following chapters are added.</p> <p>3.1.1 "HSMS Connect Mode"</p> <p>4.2.3.2 "AllPortState and PortIDList"</p> <p>10 "The Distinction from SEMI Standards"<br/>("Scenario" is shifted to Chapter11 and "Revision History" is sifted to Chapter12.)</p> <p>The followings are updated.</p> <p>4.1 "(However, when a host sends Ux item, it sends them as U2 format)" is deleted.</p> <p>4.1 The format of ALID, CEID, DATAID, ECID, RPTID, SVID, VID is altered to U4.</p> <p>4.2.2 The following contents of Table7 are altered.<br/>201 StockerAvail is deleted and to be missing number.</p> <p>5.1 S6F15/16,S6F19/20 in table9 are deleted from Message List.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|            |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|            |                  | <p>5.1 The formats related from 4.1 are altered.</p> <p>7.2.7 The following contents of the table are altered.</p> <p>501 StockerAvailChanged and 502PortStateChange are deleted and to be missing number.</p> <p>503 PortsAvailable, 504 PortsUnavailable are added.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>2.1</b> | <b>2001/1/9</b>  | <p>The followings are updated.</p> <p>1.2 "In addition, this requirement document..." and the proceeding parts are deleted.</p> <p>1.3 "Selete AMHS Integration System Emulator Requirement Document" in the Reference Documents is deleted. "Selete AMHS Communications Scenario Document" is added.</p> <p>4.2.2 Table 7 of "201 StockAvail" is added and "205 PortIDList" is deleted, "PortState" is altered to "StockerPortAvail," and "AllPortState" is altered to "AllStockerPortAvail."</p> <p>4.2.3 is entirely arranged.</p> <p>5.1 "emulator" is altered to "integrator."</p> <p>7.2.7 "501 StockerPortAvailChanged" and "502 StockerAvailChanged" are added, and "503 PortsAvailable" and "504 Ports Unavailable" are deleted.</p> <p>7.3 Repport ID 14 and 15 are updated</p> |
| <b>2.2</b> | <b>2001/2/22</b> | <p>The followings are updated.</p> <p>Use word "integrator" instead of "emulator" in whole the document.</p> <p>1.3. Reference Document Add "Selete Integrator Requirement Document" and "Selete AMHS Communication Test Specification"</p> <p>Table 7 "PortID" is altered to "StockerPortID".</p> <p>4.2.3.2 StockerPortAvail and AllStockerPortAvail "PortID" is altered to "StockerPortID".</p> <p>6.4 LOCATE 'All carrier information will be returned, when A[0] is set.' is removed from 'comment'.</p> <p>7.3. "PortID" is altered to "StockerPortID" of Report ID 14</p>                                                                                                                                                                                                          |
| <b>2.3</b> | <b>2001/8/31</b> | <p>3.2 "Variable Data Collection" in Figure2 GEM Capabilities is altered to space.</p> <p>3.2.2 "Entity that have State Model" is altered.</p> <p>Description of "Service State" and "Port State" are deleted.</p> <p>4.2.2 Table6 VID80 and 81 are added in Variable used by Stocker, which are defined by StockerSEM.</p> <p>4.2.2 Table7 "Variable used by Stocker, which are defined by Selete Specification" is deleted.</p> <p>4.2.3 "Variable defined by Selete Specifications for Stocker" is altered to Explanation of "PortTransferState and CurrentPortStates"</p> <p>7.2.2 Comment is added to CEID 51 and 52.</p> <p>7.2.7 "Event defined by Selete" is deleted.</p>                                                                                                         |

|             |                 |                                                                                                                                                                                                                                                                                                   |
|-------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             |                 | 7.2.7 "Port State Transaction events" is added.<br>7.3 Report14 is altered.<br>7.3 Report15 is deleted.<br>9.1 "Carrier Transfer to a Stocker Output Port" is deleted.<br>10 "Addiitional variables are defined by ..." is deleted.<br>10 " Additional EventReports are defined by ..."is deleted |
| <b>2.31</b> | <b>2002/2/8</b> | 5.1. Table 8, Change Examples of S1F3/4                                                                                                                                                                                                                                                           |